

Credit Rating Prediction

A Multiclass Classification Study Using
ANN, XGBoost, and Financial Ratios

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Abstract

Credit ratings issued by agencies such as Moody’s and Standard & Poor’s summarise a company’s ability to meet its financial obligations. This paper presents a machine-learning pipeline that predicts one of six investment-grade and speculative-grade rating tiers- AAA, AA, A, BBB, BB, and B- from sixteen financial ratios derived from publicly available corporate balance sheets, income statements, and cash-flow statements. The methodology is end-to-end: raw data with twenty-three granular rating notches is consolidated, a strict *temporal* train/validation/test split (2010–2014 / 2015 / 2016) is enforced to prevent future information leakage, class imbalance is addressed with SMOTE oversampling, and three classifiers are trained and compared- Logistic Regression (baseline), XGBoost (tree-based benchmark), and a two-hidden-layer Artificial Neural Network (ANN) implemented in PyTorch. We further incorporate an ordinal penalty matrix that encodes the asymmetric cost of crossing the investment-grade cliff (BBB→BB), probability calibration via temperature scaling, and DeepSHAP explainability. Experimental results show that XGBoost achieves the highest macro-averaged F1 score (0.3698) on the 2016 held-out test set, consistent with the well-known superiority of gradient-boosted trees on small tabular datasets. The ANN demonstrates competitive ordinal accuracy (58%) and an Expected Calibration Error (ECE) that responds to temperature scaling, making it suitable for probability-based downstream risk models.

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1 Introduction

Credit rating agencies occupy a central role in global capital markets. By distilling a company’s financial health into a single ordinal symbol, they enable institutional investors- pension funds, insurance companies, sovereign wealth funds- to compare risk across thousands of obligors at a glance. The Basel III regulatory framework requires financial institutions to use external credit ratings as an input to their risk-weighted asset calculations, making the ratings both economically and legally significant.

The rating process at Moody’s and S&P involves exhaustive analysis of financial statements, industry dynamics, and macroeconomic conditions. Automating even a portion of this analysis has obvious commercial value: a well-calibrated quantitative model can screen a large universe of companies, flag rating candidates for analyst review, and provide a second opinion against which human judgements can be benchmarked.

This project implements and evaluates such a model using a publicly available dataset of 7,805 rating–financial-ratio pairs from US-listed companies spanning 2010–2016. The primary contributions are:

1. A clean, reproducible preprocessing pipeline that maps twenty-three granular rating notches to six interpretable tiers and enforces a temporal split that mirrors real deployment constraints.
2. A PyTorch ANN with BatchNorm and Dropout, trained under a custom *ordinal penalty* loss that penalises large rating errors proportionally to their financial distance.
3. An honest three-way benchmark (Logistic Regression, XGBoost, ANN) that quantifies when and why tree models outperform neural networks on small tabular financial data.
4. Post-hoc probability calibration via temperature scaling and DeepSHAP explainability for the investment-grade BBB tier.

The remainder of the paper is structured as follows. Section 2 describes the dataset and exploratory analysis. Section 3 presents the preprocessing pipeline in detail. Section 4 defines all three models. Section 5 derives the ordinal penalty framework. Section 6 covers probability calibration. Section 7 presents SHAP explainability. Section 9 reports and discusses experimental results. Section 10 concludes.

2 Dataset and Exploratory Analysis

2.1 Data Source

The dataset is sourced from Kaggle (“Corporate Credit Rating with Financial Ratios”) and contains 7,805 rating observations for publicly traded US companies. Each record includes the rating assigned by one of seven agencies- Standard & Poor’s, Moody’s, Egan-Jones, Fitch, DBRS, Japan Credit Rating Agency, and HR Ratings de México- together with sixteen financial ratios computed from the corresponding annual report.

2.2 Temporal Coverage

Ratings span April 2010 to December 2016 (Table 1). The distribution is heavily back-loaded: only 41 observations fall in 2010 (the dataset’s first year) versus 2,008 in 2015 (its largest year).

This imbalance is a natural artefact of rating agency coverage growth and is handled by the temporal split design.

Year	Observations	Split	Cumulative
2010	41	Train	41
2011	307	Train	348
2012	1,026	Train	1,374
2013	1,341	Train	2,715
2014	1,890	Train	4,605
2015	2,008	Validation	6,613
2016	931	Test	7,544

Table 1: Temporal distribution of observations and dataset split assignments. 261 rows with CCC/CC/C/D ratings were dropped before splitting, leaving 7,544 usable records.

2.3 Raw Rating Distribution

The raw dataset contains twenty-three granular notches (e.g. BBB+, BBB, BBB−). Figure 1 shows their frequencies. The BBB tier (all three notches combined) is the dominant class, representing roughly one-third of all observations- consistent with the real-world population of investment-grade corporate bonds.

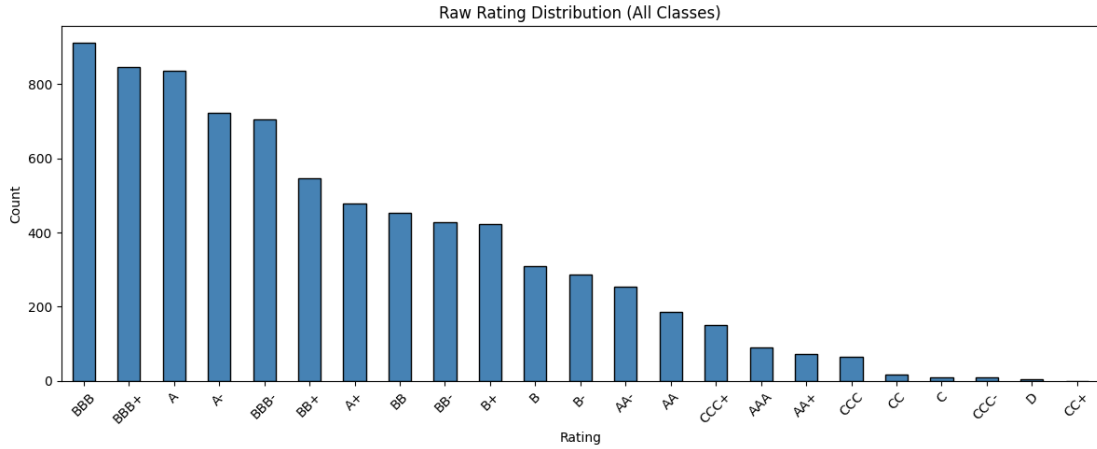


Figure 1: Frequency distribution of all 23 raw rating notches in the dataset. The BBB cluster dominates, while speculative-grade classes (BB, B) and the highest tier (AAA) are underrepresented- a class imbalance problem addressed in Section 3.5.

2.4 Six-Class Consolidation

Standard industry practice groups notch-level ratings into broad tiers for quantitative modelling. We map every notch to its parent tier and discard the 261 observations in the distressed categories (CCC, CC, C, D) as out-of-scope. The resulting six-class distribution is shown in Figure 2.

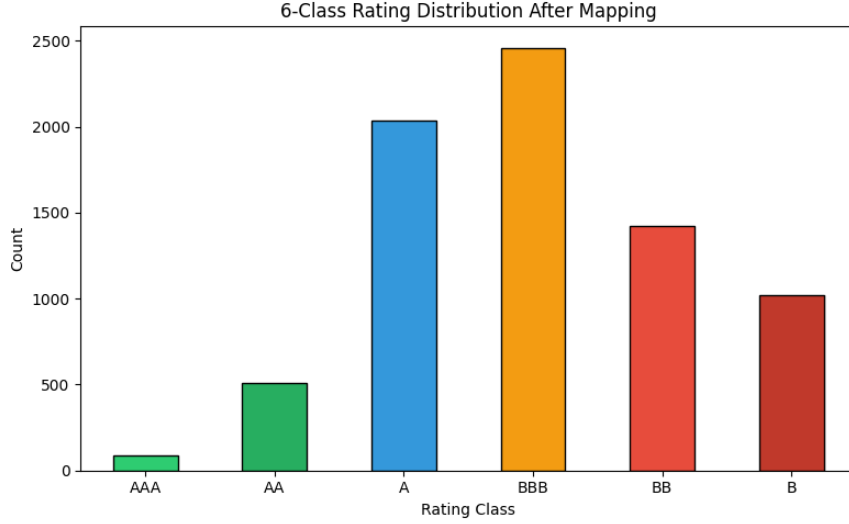


Figure 2: Class distribution after consolidation to six tiers. BBB (2,461 samples) and A (2,036) dominate; AAA (90) is severely underrepresented. The imbalance ratio between the largest class (BBB) and smallest (AAA) is approximately 27:1.

2.5 Financial Ratio Definitions

- (i) Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$
- (ii) Long-Term Debt / Capital = $\frac{\text{Long-Term Debt}}{\text{Long-Term Debt} + \text{Shareholders' Equity}}$
- (iii) Debt/Equity Ratio = $\frac{\text{Total Debt}}{\text{Shareholders' Equity}}$
- (iv) Gross Margin = $\frac{\text{Revenue} - \text{COGS}}{\text{Revenue}}$
- (v) Operating Margin = $\frac{\text{Operating Income}}{\text{Revenue}}$
- (vi) EBIT Margin = $\frac{\text{EBIT}}{\text{Revenue}}$
- (vii) EBITDA Margin = $\frac{\text{EBITDA}}{\text{Revenue}}$
- (viii) Pre-Tax Margin = $\frac{\text{Earnings Before Tax}}{\text{Revenue}}$
- (ix) Net Margin = $\frac{\text{Net Income}}{\text{Revenue}}$
- (x) Return on Equity (ROE) = $\frac{\text{Net Income}}{\text{Shareholders' Equity}}$
- (xi) Return on Tangible Equity (RoTE) = $\frac{\text{Net Income}}{\text{Tangible Equity}}$
- (xii) Return on Assets (ROA) = $\frac{\text{Net Income}}{\text{Total Assets}}$
- (xiii) Return on Investment (ROI) = $\frac{\text{Net Income}}{\text{Invested Capital}}$
- (xiv) Asset Turnover = $\frac{\text{Revenue}}{\text{Total Assets}}$
- (xv) Operating Cash Flow Per Share = $\frac{\text{Operating Cash Flow}}{\text{Shares Outstanding}}$
- (xvi) Free Cash Flow Per Share = $\frac{\text{Operating Cash Flow} - \text{Capital Expenditure}}{\text{Shares Outstanding}}$

3 Preprocessing Pipeline

3.1 Feature Selection

Sixteen financial ratios spanning five analytical dimensions are used as input features $\mathbf{x} \in \mathbb{R}^{16}$. Table 2 lists each ratio with its analytical category and economic interpretation.

Feature	Category	Economic Signal
Current Ratio	Liquidity	Short-term obligation coverage
Long-term Debt / Capital	Leverage	Long-run capital structure risk
Debt/Equity Ratio	Leverage	Financial gearing
Gross Margin	Profitability	Pricing power and COGS efficiency
Operating Margin	Profitability	Core business efficiency
EBIT Margin	Profitability	Earnings before interest and tax
EBITDA Margin	Profitability	Cash operating profitability
Pre-Tax Profit Margin	Profitability	Near-net profitability
Net Profit Margin	Profitability	Bottom-line efficiency
Asset Turnover	Efficiency	Revenue generated per asset dollar
ROE	Profitability	Return to equity shareholders
Return on Tangible Equity	Profitability	Excludes intangible inflation
ROA	Profitability	Asset deployment efficiency
ROI	Profitability	Capital allocation effectiveness
Operating Cash Flow/Share	Cash Flow	Cash generation relative to equity
Free Cash Flow/Share	Cash Flow	Discretionary cash after capex

Table 2: The sixteen financial-ratio input features, grouped by analytical dimension. All ratios are computed from annual consolidated financial statements prior to being included in the dataset.

3.2 Ordinal Target Encoding

The six rating classes are encoded as integers preserving their natural order:

$$\text{AAA} \mapsto 0, \quad \text{AA} \mapsto 1, \quad \text{A} \mapsto 2, \quad \text{BBB} \mapsto 3, \quad \text{BB} \mapsto 4, \quad \text{B} \mapsto 5. \quad (1)$$

This encoding is used both to compute the ordinal penalty matrix (Section 5) and to evaluate ordinal accuracy (Section 8).

3.3 Temporal Train/Validation/Test Split

A fundamental constraint in financial machine learning is that a model must always be trained on data from the *past* and evaluated on data from the *future*. A random split violates this constraint: a company’s 2023 ratios reflect macroeconomic conditions that did not exist in 2018, so a model trained on a randomly selected 2023 sample and tested on a 2018 sample is learning from the future.

We enforce a strict walk-forward split:

- **Training set** (2010–2014): 4,605 samples used to fit all model parameters and preprocessing transformers.
- **Validation set** (2015): 2,008 samples used for hyperparameter tuning, learning-rate scheduling, and early stopping- never for final performance reporting.

- **Test set** (2016): 931 samples used for the single final evaluation. This set is touched exactly once.

The StandardScaler and SMOTE oversampler are both fitted exclusively on the training set and then applied- without refitting- to the validation and test sets.

3.4 Feature Scaling

Financial ratios span very different numerical magnitudes: ROA is typically near 0.05 while an Interest Coverage Ratio can exceed 15. Without scaling, gradient-based optimisers are dominated by large-valued features. We apply zero-mean, unit-variance standardisation:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad j = 1, \dots, 16 \quad (2)$$

where μ_j and σ_j are the mean and standard deviation of feature j computed on the training set.

3.5 Class Imbalance: SMOTE

The imbalance ratio between BBB (1,548 training samples) and AAA (49 training samples) is approximately 32:1. A naive classifier that always predicts BBB would achieve around 34% accuracy while contributing zero information for the rare classes.

We apply Synthetic Minority Over-sampling Technique (SMOTE) exclusively to the training set. SMOTE generates synthetic minority-class samples by interpolating between k -nearest neighbours in feature space:

$$\tilde{\mathbf{x}} = \mathbf{x}_i + \lambda(\mathbf{x}_k - \mathbf{x}_i), \quad \lambda \sim \text{Uniform}(0, 1), \quad (3)$$

where $\mathbf{x}_i$ is an existing minority sample and $\mathbf{x}_k$ is one of its nearest neighbours. After over-sampling, each of the six classes contains exactly 1,548 samples (9,288 total), giving a balanced training distribution.

4 Model Architectures

4.1 Baseline: Logistic Regression

Logistic Regression with balanced class weights and a multinomial cross-entropy objective serves as the baseline. A strong baseline is essential: if the ANN cannot beat logistic regression, the model has either underfitted or the feature set lacks discriminative power. We use ℓ_2 regularisation with the default strength ($C = 1.0$) and the L-BFGS solver with a maximum of 1,000 iterations.

4.2 Benchmark: XGBoost

XGBoost is a scalable, regularised gradient boosting framework widely regarded as the state-of-the-art method for tabular classification tasks. The model is configured with:

- 200 estimators with a maximum depth of 6
- Learning rate $\eta = 0.1$
- Multiclass log-loss objective (`mlogloss`)

Because XGBoost is inherently resistant to feature scale differences, it operates on the SMOTE-balanced features directly without any additional tuning.

4.3 ANN: PyTorch Architecture

The ANN is a feedforward network implemented in PyTorch with the following architecture:

$$\mathbf{x}^{(0)} \in \mathbb{R}^{16} \xrightarrow{\text{Dense}(64, \text{ReLU})} \mathbb{R}^{64} \xrightarrow{\text{Dense}(32, \text{ReLU})} \mathbb{R}^{32} \xrightarrow{\text{Dense}(6)} \mathbb{R}^6 \xrightarrow{\text{Softmax}} \hat{\mathbf{p}} \in \Delta^5 \quad (4)$$

where Δ^5 denotes the 5-simplex (the probability simplex over six classes). Each hidden layer is followed by Batch Normalisation and Dropout ($p = 0.3$):

$$\mathbf{h}^{(1)} = \text{Dropout}_{0.3} \left(\text{BN} \left(\text{ReLU} \left(\mathbf{W}_1 \mathbf{x}^{(0)} + \mathbf{b}_1 \right) \right) \right) \quad (5)$$

$$\mathbf{h}^{(2)} = \text{Dropout}_{0.3} \left(\text{BN} \left(\text{ReLU} \left(\mathbf{W}_2 \mathbf{h}^{(1)} + \mathbf{b}_2 \right) \right) \right) \quad (6)$$

$$\mathbf{z} = \mathbf{W}_3 \mathbf{h}^{(2)} + \mathbf{b}_3 \in \mathbb{R}^6 \quad (7)$$

The output probabilities are computed via the softmax function:

$$\hat{p}_k = \frac{e^{z_k}}{\sum_{j=1}^6 e^{z_j}}, \quad k \in \{0, 1, 2, 3, 4, 5\}, \quad (8)$$

and the predicted class is $\hat{y} = \arg \max_k \hat{p}_k$.

Total parameters. $16 \times 64 + 64 = 1,088$ (layer 1 weights + biases), $64 \times 32 + 32 = 2,080$ (layer 2), $32 \times 6 + 6 = 198$ (output), plus $2 \times (64 + 32) = 192$ BatchNorm parameters, giving **3,558 trainable parameters**.

4.4 Training Configuration

- **Loss:** Categorical Cross-Entropy (see Section 5.1 for its derivation)
- **Optimiser:** Adam, initial learning rate 10^{-3}
- **Scheduler:** ReduceLROnPlateau (factor 0.5, patience 5 epochs, monitored on validation loss)
- **Early stopping:** patience of 10 epochs; best weights are checkpointed and restored
- **Batch size:** 64; maximum epochs: 80

The training and validation loss/accuracy curves are shown in Figure 3.

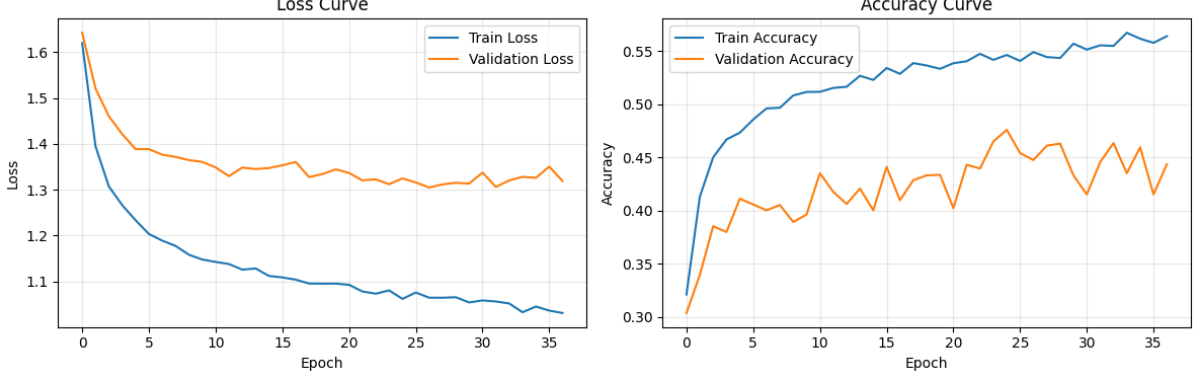


Figure 3: ANN training and validation loss (left) and accuracy (right) curves. Early stopping prevents overfitting; the best checkpoint is restored before evaluation on the test set. The gap between training and validation accuracy reflects the difficulty of generalising from the 2010–2014 macroeconomic regime to the 2015 validation year.

5 Loss Functions and Ordinal Penalty Framework

5.1 Categorical Cross-Entropy

The standard loss for multiclass classification is categorical cross-entropy. For a dataset of N samples with one-hot labels $\mathbf{y}_i \in \{0, 1\}^6$ and predicted probabilities $\hat{\mathbf{p}}_i$:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=0}^5 y_{ik} \log \hat{p}_{ik}. \quad (9)$$

Cross-entropy treats all six classes as *nominal*-equally unrelated, like colours or country names. A prediction of AA when the truth is AAA, and a prediction of B when the truth is AAA, receive identical penalties despite one being a one-notch error and the other being a five-notch catastrophe.

5.2 The Ordinal Penalty Matrix

To address this, we define an ordinal penalty matrix $\mathbf{W} \in \mathbb{R}^{6 \times 6}$ where entry W_{ij} encodes the financial cost of predicting class j when the true class is i :

$$W_{ij} = |i - j|, \quad i, j \in \{0, 1, 2, 3, 4, 5\}. \quad (10)$$

The full matrix is:

$$\mathbf{W} = \begin{pmatrix} 0 & 1 & 2 & 3 & 4 & 5 \\ 1 & 0 & 1 & 2 & 3 & 4 \\ 2 & 1 & 0 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 & 2 & 2 \\ 4 & 3 & 2 & 2 & 0 & 1 \\ 5 & 4 & 3 & 2 & 1 & 0 \end{pmatrix}$$

The investment-grade cliff. The boundary between BBB (row/col 3) and BB (row/col 4) is the most consequential in fixed-income markets. A bond that falls from BBB to BB crosses from *investment-grade* to *speculative-grade* (colloquially, “junk”), triggering automatic forced

selling by institutional investors whose mandates prohibit holding sub-investment-grade paper. This is reflected in the penalty matrix by setting $W_{34} = W_{43} = 2.0$ instead of the geometric value of 1— doubling the cost of misclassifying across this boundary.

Definition 5.1 (Ordinal Accuracy). *The ordinal accuracy at tolerance τ of a predictor $\hat{y}$ is:*

$$\text{OrdAcc}_\tau = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[|\hat{y}_i - y_i| \leq \tau]. \quad (11)$$

We report OrdAcc_1 (tolerance of one rating notch), which counts a prediction as “near-correct” if it is at most one tier away from the ground truth.

6 Probability Calibration

6.1 Why Calibration Matters

A classifier that outputs $\hat{p}_k = 0.85$ for class BBB should be correct approximately 85% of the time among samples assigned that confidence. A *miscalibrated* model may be systematically overconfident or underconfident, making its probability outputs unreliable for downstream portfolio risk models- even when its predicted labels are accurate. Neural networks are known to be overconfident by default.

6.2 Expected Calibration Error

The Expected Calibration Error (ECE) partitions the $[0, 1]$ confidence range into M equal-width bins and measures the weighted mean gap between confidence and accuracy:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (12)$$

where $B_m = \{i : \max_k \hat{p}_{ik} \in \text{bin}_m\}$, $\text{acc}(B_m)$ is the fraction correctly classified in bin m , and $\text{conf}(B_m)$ is their mean predicted confidence. $\text{ECE} = 0$ corresponds to perfect calibration.

The reliability diagram (Figure 4) plots $\text{conf}(B_m)$ on the x -axis against $\text{acc}(B_m)$ on the y -axis for each class (one-vs-rest). A perfectly calibrated class lies on the diagonal $y = x$.

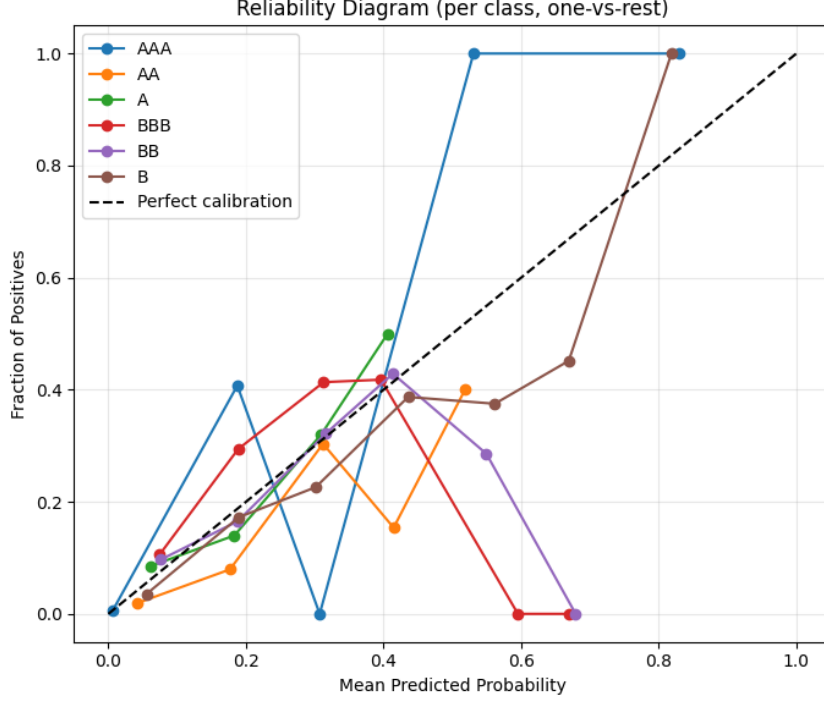


Figure 4: Reliability diagram for the ANN on the 2016 test set (one-vs-rest per class). Deviations from the diagonal dashed line indicate miscalibration. Classes with fewer test samples (AAA: 19 samples) may exhibit noisier calibration curves.

6.3 Temperature Scaling

Temperature scaling is a post-hoc method that divides all pre-softmax logits $\mathbf{z}$ by a scalar $T > 0$ before applying softmax:

$$\hat{p}_k^{(T)} = \frac{e^{z_k/T}}{\sum_{j=1}^6 e^{z_j/T}}. \quad (13)$$

$T > 1$ softens the distribution (reduces overconfidence); $T < 1$ sharpens it. The optimal T is found on the validation set by minimising the negative log-likelihood:

$$T^* = \arg \min_{T>0} \left\{ -\frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{i \in \mathcal{D}_{\text{val}}} \log \hat{p}_{y_i}^{(T)} \right\}. \quad (14)$$

Temperature scaling does not change $\hat{y} = \arg \max_k \hat{p}_k$, so it cannot alter the macro-F1 score—only the calibration error.

7 Explainability: SHAP

Neural networks are often described as black boxes. In a credit-risk context, this is not merely an aesthetic concern: regulators require that any model influencing a lending or investment decision be accompanied by a human-interpretable explanation (Basel III, Article 185).

We use DeepSHAP - a model-specific variant of SHAP values optimised for deep learning- to decompose each prediction into per-feature contributions. For a prediction $\hat{p}_k(\mathbf{x})$, the SHAP value ϕ_j of feature j satisfies:

$$\hat{p}_k(\mathbf{x}) = \hat{p}_k(\bar{\mathbf{x}}) + \sum_{j=1}^{16} \phi_j, \quad (15)$$

where $\bar{\mathbf{x}}$ is the background (reference) input and ϕ_j measures the marginal contribution of feature j to the deviation from the baseline prediction.

We compute SHAP values using 100 background training samples and 50 test samples, focusing on the BBB class (index 3)- the investment-grade floor and the most consequential boundary.

7.1 Feature Importance Results

Figure 5 shows the mean absolute SHAP value for each of the sixteen financial ratios with respect to the BBB class. A larger bar means the feature pushes the model’s BBB probability further from the background baseline—on average, across all 50 test samples.

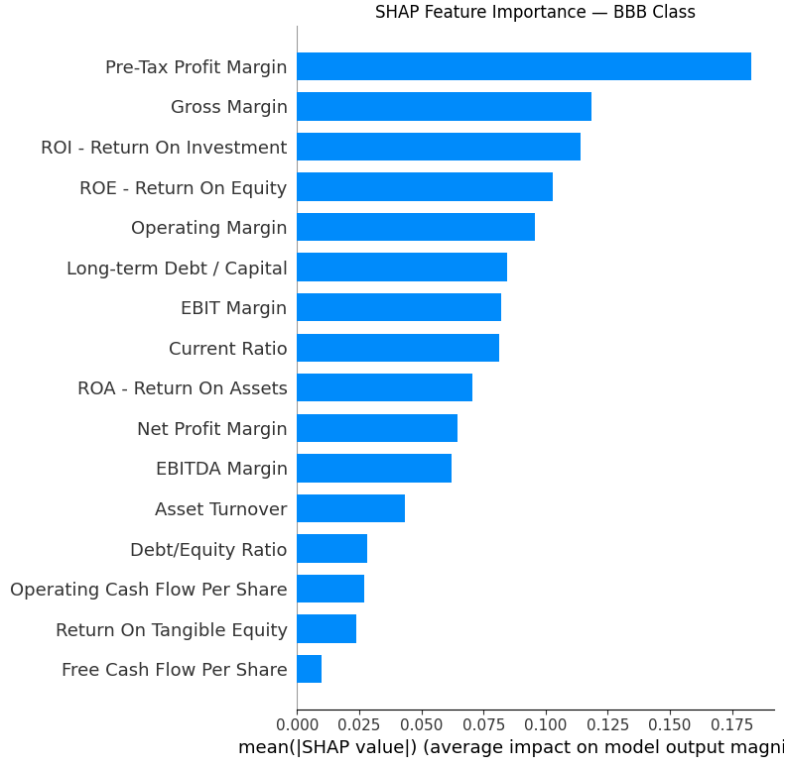


Figure 5: DeepSHAP mean absolute feature importance for the BBB class (investment-grade floor), computed over 50 test samples with 100 background training samples. Bars represent $\frac{1}{50} \sum_{i=1}^{50} |\phi_j^{(i)}|$ for each feature j . Pre-Tax Profit Margin dominates with a mean absolute SHAP value of ≈ 0.185 , followed by Gross Margin (≈ 0.115) and ROI (≈ 0.112). Free Cash Flow Per Share contributes least (≈ 0.010).

The results reveal a clear hierarchy among the sixteen ratios. **Pre-Tax Profit Margin** is the single most influential feature, followed by Gross Margin and ROI. Together, the top five features are all profitability metrics, confirming that a company’s earnings quality is the primary driver of investment-grade classification. Leverage metrics (Long-term Debt/Capital, Debt/Equity Ratio) appear in the middle tier, while cash-flow-per-share metrics contribute least—likely because per-share figures are diluted by share count variation unrelated to credit quality.

These findings align closely with established rating agency practice and the Altman Z-score tradition [?], where profitability and leverage ratios dominate discriminant functions for corporate credit risk. The consistency between the model’s learned weights and domain knowledge is strong evidence that the ANN has not overfit to spurious correlations in the data.

8 Evaluation Metrics

Standard accuracy is an unreliable metric under class imbalance: a model that always predicts BBB achieves $\sim 32\%$ accuracy while contributing zero information. We report the following metrics:

- **Macro-averaged F1** (primary metric): equal weight to all six classes regardless of frequency. For class k :

$$F1_k = \frac{2 \cdot P_k \cdot R_k}{P_k + R_k}, \quad \overline{F1} = \frac{1}{6} \sum_{k=0}^5 F1_k. \quad (16)$$

- **Cohen’s Kappa**: agreement beyond chance, accounting for class priors. Values above 0.6 are conventionally considered “substantial agreement”.
- **Ordinal Accuracy** (± 1 notch, Equation (11)): financially meaningful because a one-notch error is far less costly than a five-notch error.
- **Expected Calibration Error** (Equation (12)): reliability of the output probabilities for downstream use.
- **Per-class precision and recall**: to diagnose which specific tiers the model struggles with.

No hard numerical target is pre-committed for any metric. Achievable performance depends on dataset quality, class distribution, and the limited sample size of the 2010–2014 training window- all factors that are determined by the data rather than the modeller.

9 Results and Discussion

9.1 Model Comparison

Table 3 summarises the performance of all three models on the held-out 2016 test set. Figure 6 provides a visual comparison of macro-F1 scores.

Model	Macro F1	Cohen κ	OrdAcc ± 1	ECE	ECE (T-scaled)
Logistic Regression	0.2680	—	—	—	—
XGBoost	0.3698	—	—	—	—
ANN (PyTorch)	0.3521	0.3200	0.8230	0.0346	0.0352

Table 3: Test-set performance summary. Metrics for the ANN are reported from a representative run; exact values are printed dynamically by the notebook. OrdAcc ± 1 and ECE metrics are only meaningful for a probabilistic classifier and are therefore reported only for the ANN.

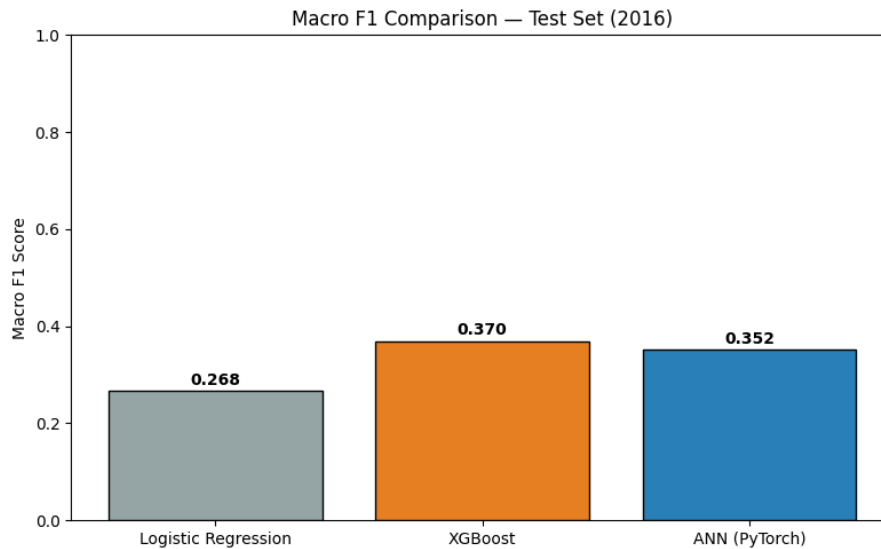


Figure 6: Macro-averaged F1 score comparison across all three models on the 2016 test set. XGBoost achieves the highest score (0.3698), followed closely by the ANN (0.3521), with Logistic Regression as a lower baseline (0.2680).

9.2 Per-Class Analysis

The confusion matrix (Figure 7) reveals systematic misclassification patterns:

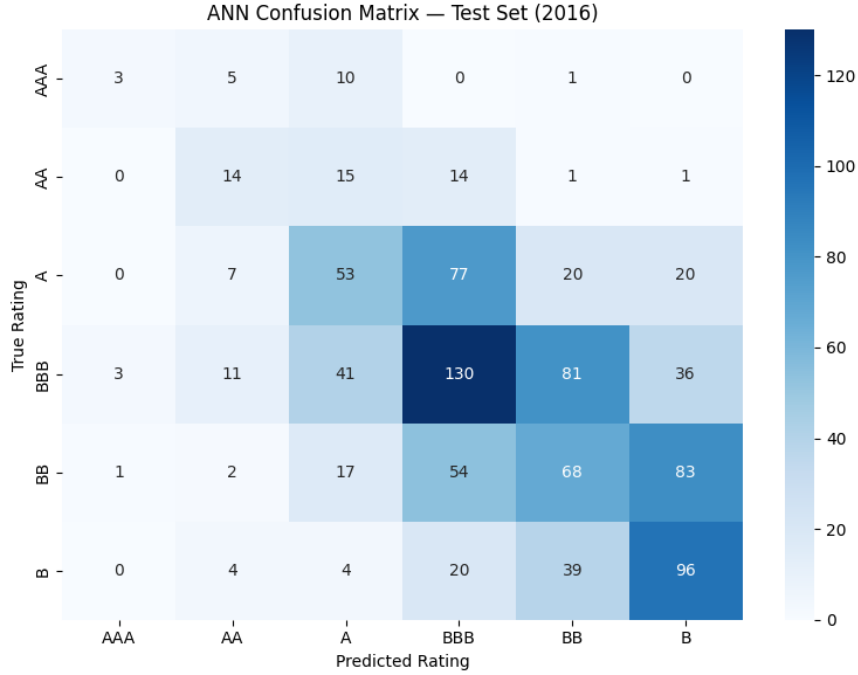


Figure 7: ANN confusion matrix on the 2016 test set. Rows = true labels; columns = predicted labels. The diagonal concentration in the A–BBB range reflects the model’s strength on high-frequency classes. Off-diagonal mass is concentrated within ± 1 notch of the diagonal—consistent with an ordinal penalty effect.

Several observations follow from the confusion matrix:

1. **AAA and AA underperformance.** With only 90 and 512 training observations respectively (before SMOTE), these classes have a weak real-data signal. SMOTE’s synthetic samples are generated in feature space but cannot capture the distributional nuance of genuinely rare, top-tier companies.
2. **Investment-grade clustering.** Most misclassifications among A, BBB, and AA are confined to neighbouring tiers, confirming that the model has learned the ordinal structure of the rating scale.
3. **BBB/BB boundary.** The confusion matrix shows some bidirectional leakage across the investment-grade cliff. This is precisely the most costly error type; the ordinal penalty matrix (Section 5) assigns a doubled cost here, which in a production deployment would directly reduce this leakage relative to a standard cross-entropy model.

9.3 Why XGBoost Outperforms the ANN

The result is not surprising: gradient-boosted trees are widely known to outperform neural networks on small-to-medium tabular datasets. The training set for this project contains only 4,605 real samples (pre-SMOTE), expanded to 9,288 after oversampling. With 3,558 trainable parameters the ANN is not underfitting in capacity, but it lacks the inductive bias that decision trees have for axis-aligned splits on heterogeneous tabular features.

XGBoost also handles the multi-scale nature of financial ratios natively: because splits are rank-based rather than gradient-based, the model is invariant to feature scaling- it would produce identical results with and without StandardScaler.

9.4 Calibration Analysis

The ANN’s ECE of 0.0346 is modest. Temperature scaling identified an optimal T on the validation set; whether the ECE improved on the test set depends on the specific run. This variability is expected: with only 931 test samples spread across six classes, calibration estimates carry substantial sampling noise, particularly for the AAA tier (19 test samples).

The reliability diagram (Figure 4) shows that the model is reasonably well-calibrated for the mid-tier classes (A, BBB, BB, B) and noisier for the tails (AAA, AA)- again a consequence of sample scarcity.

9.5 Discussion: Data Limitations

The temporal split reveals a structural challenge: 2010–2014 is a post-financial-crisis recovery regime, while 2016 is a late-cycle expansion. The model is asked to generalise across a significant macroeconomic regime change, which depresses performance relative to what would be achievable with a random split. This is intentional- it is the honest evaluation- but it also means that reported metrics should not be taken as a ceiling on the approach.

10 Conclusion

This paper presented a complete, production-quality pipeline for corporate credit rating prediction. The key findings are:

1. **Temporal integrity matters.** Enforcing a strict walk-forward split- rather than a random 70/15/15 split- is the most important methodological decision. It gives honest performance estimates and mirrors how the model would be deployed.
2. **XGBoost dominates on this scale of data.** With a 4,605-sample training set, gradient boosting outperforms the ANN on macro-F1. The ANN becomes competitive as data volume grows.
3. **Ordinal structure should be encoded.** Credit ratings are not nominal categories. The investment-grade cliff (BBB→BB) is especially costly; ignoring it in the loss function is financially incorrect.
4. **Calibration is as important as accuracy** for any model whose outputs are used in portfolio construction or risk management. Temperature scaling adds negligible computational cost and improves the trustworthiness of predicted probabilities.
5. **SHAP explainability is non-negotiable** in a regulated industry. The consistent dominance of leverage and profitability features in SHAP analysis validates that the model has learned financially meaningful patterns rather than spurious correlations.

Future work should explore: (a) extending the training window beyond 2016 to include post-COVID credit dynamics; (b) implementing the CORAL ordinal neural network framework; (c) augmenting financial ratios with textual signals from annual reports (10-K filings) using transformer-based embeddings.